

Lecture 18 — Matrices as data and structured objects

Mon Oct 5

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Related reading and the course-wide list of daily objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.